

COMPUTER VISION TRACK

AI Engineer Roadmap: Curated YouTube Playlists

4 free, hand-picked playlists that together cover every topic in the Computer Vision roadmap — from OpenCV basics to model optimization for on-device Reels.

No single playlist covers the full roadmap in depth. The four below combine practical OpenCV, deep-learning fundamentals, real-world detection/segmentation projects, and deployment/optimization — all free on YouTube.

1

OpenCV Course — Full Tutorial with Python

freeCodeCamp.org | youtube.com/watch?v=oXlwWbU8I2o

A complete, beginner-friendly walkthrough of OpenCV fundamentals — the foundation every CV pipeline is built on.

WHAT IT COVERS

- ✓ Reading & processing images/video, resizing and rescaling
- ✓ Image transformations, color spaces, and edge detection
- ✓ Contour detection, bitwise operations, masking, histograms, thresholding
- ✓ Face detection with Haar Cascades and OpenCV face recognition
- ✓ Capstone: a deep computer-vision classifier project

Why it matters for Reels: Gives you the raw image-processing toolkit needed to decode, clean, and enhance every video frame before any model ever sees it.

2

PyTorch for Deep Learning & Machine Learning — Full Course

freeCodeCamp.org (Daniel Bourke) | youtube.com/watch?v=V_xro1bcAuA

A 25-hour, code-first course that builds up from tensors to full convolutional neural networks in PyTorch.

WHAT IT COVERS

- ✓ PyTorch fundamentals: tensors, autograd, training/testing loops, GPU workflow
- ✓ Neural network classification from first principles
- ✓ Computer vision with CNNs: torchvision, Conv2d/MaxPool2d, training on image data
- ✓ Custom datasets end-to-end (loading, augmenting, training, evaluating)
- ✓ Confusion matrices and practical model-evaluation techniques

Why it matters for Reels: Builds the CNN and PyTorch skills you'll reuse for every architecture in the roadmap — ResNet, EfficientNet, ViT, YOLO, and beyond.

More Curated Playlists

3

Augmented Startups — Computer Vision Playlists

youtube.com/@AugmentedStartups/playlists

Project-driven playlists focused on the newest detection, segmentation, and tracking models, built for shipping real applications.

WHAT IT COVERS

- ✓ YOLO (latest versions) for real-time object detection
- ✓ OpenCV-based practical pipelines
- ✓ Segment Anything (SAM) for zero-shot segmentation
- ✓ Face detection & face recognition projects
- ✓ Multi-object tracking and end-to-end deployment

Why it matters for Reels: Turns theory into shippable Reels features — product detection, AR background separation, and live tracking, the way production teams actually build it.

4

LearnOpenCV — Full Channel Playlists

youtube.com/@LearnOpenCV/playlists

Content straight from the OpenCV.org team, spanning classical CV through today's transformer-based and multimodal models.

WHAT IT COVERS

- ✓ OpenCV fundamentals and image processing
- ✓ DETR and modern transformer-based detectors
- ✓ Segment Anything and Vision Transformers (ViT)
- ✓ CLIP and vision-language / multimodal models
- ✓ ONNX, TensorRT, quantization & pruning for deployment

Why it matters for Reels: Closes the loop on the roadmap's hardest part — getting cutting-edge models like CLIP and ViT running fast, on-device, in real time.

Topic Coverage Matrix

Every topic from the roadmap, mapped to which playlist(s) teach it.

Topic	Covered by playlist(s)
OpenCV	1, 3, 4
Image processing / filters / augmentation	1, 2

Topic	Covered by playlist(s)
CNNs	2
ResNet / EfficientNet	2
Vision Transformer (ViT)	4
YOLO	3
Faster R-CNN / DETR	4
U-Net / Mask R-CNN	3
Segment Anything (SAM)	3, 4
Face detection & recognition	1, 3
Facial landmarks	3
Video understanding / action recognition	2, 3
3D CNNs	2
CLIP / vision-language models	4
ONNX / TensorRT	4
Quantization & pruning	4

Legend: 1 = OpenCV Course (freeCodeCamp) · 2 = PyTorch for Deep Learning (freeCodeCamp) · 3 = Augmented Startups · 4 = LearnOpenCV

Suggested order: Start with Playlist 1 (OpenCV basics) → Playlist 2 (CNNs & PyTorch) → Playlist 3 (YOLO, segmentation, real projects) → Playlist 4 (transformers, CLIP, deployment).